

Malhar Gudekar

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EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Data Warehousing & Business Intelligence, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: DBMS, Big Data Analysis, Data Warehousing and Mining, Project Management, NLP, Machine Learning, Social Media Analytics

SKILLS

• **Analytical:** Data Analysis, Exploratory Data Analysis, Statistical Analysis, Trend Analysis, Root Cause Analysis, KPI Reporting, Data Validation, Data Cleaning, Anomaly Detection, ETL, Predictive Analytics

• **Programming / Querying:** Python, SQL, Pandas, NumPy, PySpark, PostgreSQL, Neo4j

• **Visualization / Reporting:** Power BI, Tableau, DAX, Excel, VBA, Dashboard Development, Data Visualization, SQL Server, Power BI Service, Power BI Desktop, Star Schema Modeling, SSMS

• **Data / Cloud / Tools:** AWS S3, EC2, Redshift, EKS, FastAPI, Docker, Git, Kafka, pgVector, GCP, Azure Data Factory

WORK EXPERIENCE

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

• Designed and maintained relational data models and ingestion workflows for a cross-platform system tracking 100+ users, enabling structured reporting and downstream analytics consumption

• Designed and optimized **PostgreSQL architecture** for high-volume, production-scale datasets, improving query performance by 38%

• Built robust, production-ready **Python** pipeline with retrieval and context summarization, improving multi-turn response accuracy by 43%

• Developed **backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on reliability and scalability

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

• Designed and deployed highly **scalable NLP pipelines in Python** to process high-volume datasets, reducing processing latency by 41%

• Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing and pattern detection

• Developed production-grade **APIs** and deployment workflows to improve system reliability and reproducibility

• Enabled large-scale data processing across social media streams using **distributed system architecture** principles

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

• Supported development of Power BI reporting solutions by designing ETL workflows to extract and transform data from multiple enterprise sources into structured, analytics-ready datasets

• Documented data models, naming conventions, and pipeline logic to align reporting outputs with source system definitions and support long-term report maintainability

• Improved system performance by 64% through optimized indexing strategies and pipeline efficiency improvements

• Developed backend services enabling real-time, high-performance querying of unstructured data

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

• Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by 40% and improving visibility into operational performance across transportation services

• Processed and optimized large-scale operational datasets using **Python** and **PySpark** to identify failure, trends, and performance issues

• Wrote and optimized SQL queries against relational databases to extract, validate, and reconcile operational data from enterprise sources, resolving data quality issues and speeding root-cause analysis by 26%

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

• Developed and maintained 20+ Power BI reports and dashboards for enterprise clients across Sales, Finance, and Operations, building semantic models with DAX measures, calculated columns, and table relationships to enable KPI tracking across business units

• Validated and reconciled reporting data across enterprise datasets using SQL and Excel, standardizing definitions and following documentation practices that cut report preparation time by 42% and improved consistency across weekly cycles

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

• Led cross-functional team of 6 to deliver product updates for DeepCover, game, using user feedback and performance insights to guide data-driven feature decisions

• Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, improving cross-team visibility and supporting on-time milestone reviews

PROJECTS

FMCG Sales Demand Forecasting

March 2025

[View Project](#)

• Analyzed FMCG **sales demand data** across categories and locations to identify trends, seasonal patterns, and key drivers impacting inventory

• Built and compared forecasting models including **ARIMA**, **SARIMA**, and regression to evaluate demand patterns and improve forecast accuracy using RMSE and MAPE

• Derived actionable insights on promotions, pricing, and stock behavior, enabling better inventory planning and reducing stockouts